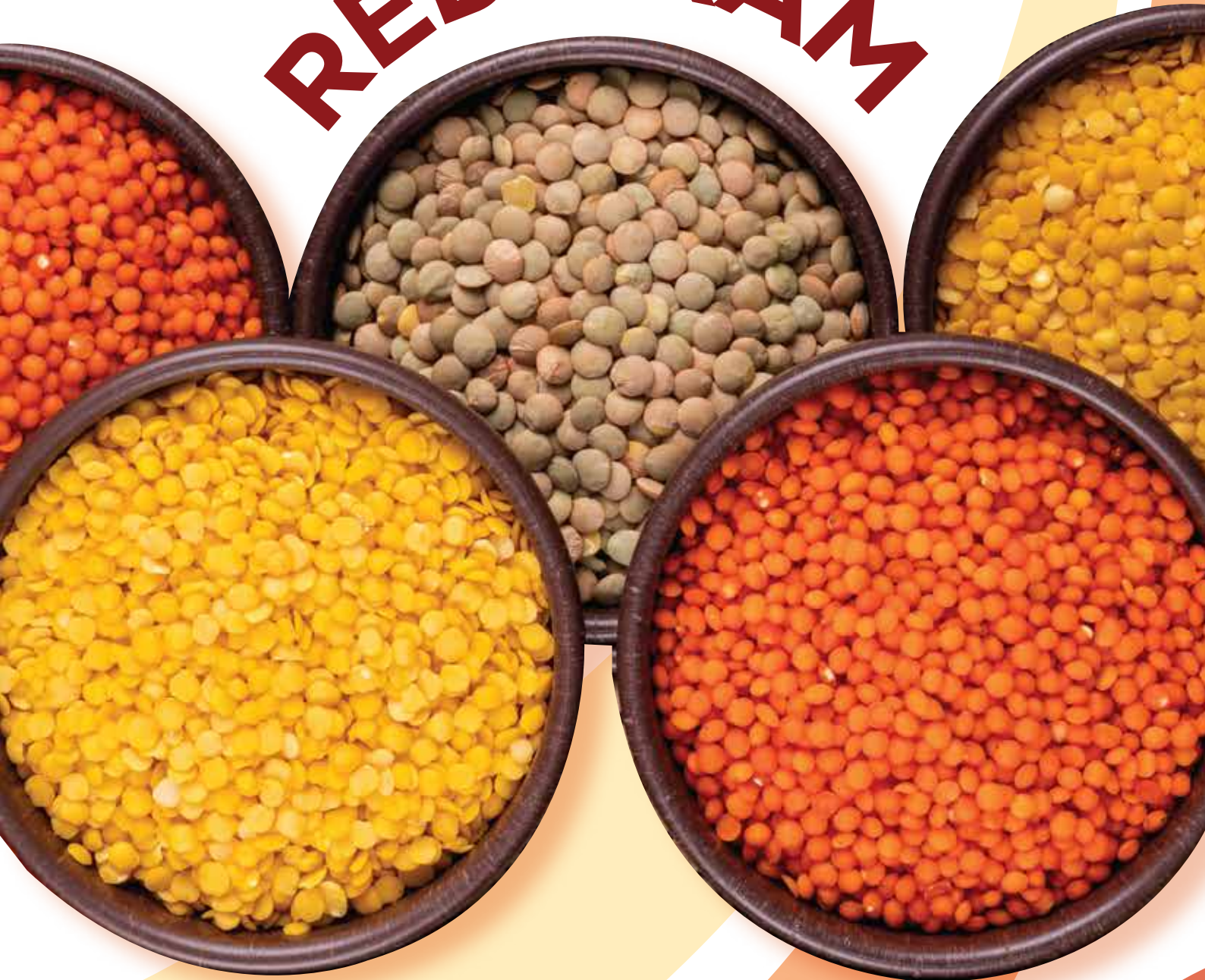




RED GRAM



Visvesvaraya Trade Promotion Centre (VTPC)

VTPC Building, Kasturba Road, Bengaluru-560001

Preface

With a robust agro potential the Country has, the Government of India has launched the One District One Focus Product Scheme (ODOFP) for agriculture sector by the Ministry of Food Processing Industries. Among its primary objectives, the ODOFP initiative aims to enhance the value of the products which is eventually expected to boost employment and income levels of the farmers. The identified products across the country is available at <https://www.nfsm.gov.in/odopstatecropsreport.aspx>

Karnataka has identified its product mix with high demand and export potential across all the districts. The implementing agency for this initiative is the Karnataka State Agricultural Produce Processing and Export Corporation Limited (KAPPEC), the nodal agency in the State.

VTPC being the nodal agency for promotion of exports from the State has made an attempt to draw the action plan district wise, to capture the potential, present status and future prospects in domestic and international markets. Besides, detailed insights have been provided into the biological description of the product, their local, national and international varieties, export-import analysis, HS codes. For a holistic perspective for those concerned, each report also provides SPS standards, processing technologies available, export grading and packing specifications, and suggested pivotal roles and responsibilities among the government departments, boards, corporations and Universities.

Presenting the Action Plan/Report for Red Gram , an ODOFP product mapped to Kalburgi district of Karnataka, formulated by VTPC. I sincerely hope that this ready reckoner with first-hand information regarding the farming sector would help all those concerned, especially the FPOs and food processing entrepreneurs who have a desire to focus on exporting this product.

S.R Satheesha

Director (Exports) & Managing Director

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Introduction

The pigeon pea botanically called *Cajanus cajan*. (L) Millsp., is a perennial legume from the family Fabaceae. It is a drought-tolerant, perennial, but mostly cultivated as an annual, erect shrub. It grows well in a wide range of soil types, varying from sandy to heavy loams. Small, rust-brown to yellow flowers appear in racemes, which develop into elongated fruit pods, enclosing 2 to 8 seeds separated from each other in the pod by slight depressions.

The pods, like in other legumes, opens along a seam on either side. Raw green seeds are ovoid, measure 6-8 mm across which turn brown, red, patterned-black as they dry depending on the cultivar type.

In India, pigeon pea is split and sold as red gram, which is a popular pulse crop. Arhar and Tur are some of the other names for it. It is consumed on a large scale in South Asia and is a major source of protein for the population of the Indian subcontinent. It is the primary accompaniment to rice or roti (flat bread) and has the status of staple food throughout the length and breadth of India. Other than India, red gram is mostly grown and consumed in underdeveloped nations. Pigeon peas are small, round to oval, mostly cherry-red coloured dicotyledonous seeds in the legume family. They do, however, come in a variety of colour patterns, such as yellow, brown, variegated, and so on. Since its domestication in the Indian subcontinent at least 3,500 years ago, it is now farmed as a significant pulse crop throughout Asia's tropics and subtropics, as well as Sub-Saharan Africa and Latin America, for its nutritious dry seeds.

Every Red gram plant is a mini-fertilizer factory as the crop has unique characteristics of restoring and maintaining soil fertility through fixing atmospheric nitrogen in symbiotic association with Rhizobium bacteria present in the root nodules. Red gram crop is suitable for inter-cropping, with different crops (Cotton, Sorghum, Pearl millet, Green gram, Black gram, Maize, Soybean, Groundnut) for increasing production and maintaining soil fertility.

For more details on Tur dal farming, planting, care, harvesting and yield, please visit:

<https://www.agrifarming.in/tur-dal-farming>

Health benefits of red gram: <https://www.nutrition-and-you.com/pigeon-peas.html>

Red gram varieties cultivated in India

Varieties	Parentage	Year of release	Duration (days)	Grain yield (Kg/ha)		Special features
				Rain fed	Irrigated	
Vamban 2	ICPL 341 x Bhavani sagar local	1999	170-180	1050	-----	Suited for entire Tamil Nadu to replace SA 1 & CO6 S.O. 821 (E) /13.09.2000
CO(RG) 7	Selection from PB 9825	2004	120-130	950	1170	Bold seeded and suitable for late onset of monsoon S.O. 1177 (E) / 25.08.2005
VCN(Rg)3	VCN 1 x Gulbarga	2005	100-105	884	1530	Determinate, high yielding, short duration S.O. 821 (E) /13.09.2006
CO 8	APK 1 x LRG 41	2017	170-180	1600	1800	Sterility mosaic disease resistant and root rot S.O.1379 (E)/27.03.2018
CO 9 (CRG 2012-25)	CO 6 x IC 525427	2019	170-180	1700	-----	Bold seed (100- seed weight 9.9 g) Moderately resistant to Wilt and SMD diseases.

Other varieties of India - National Pulses Research Centre, Vamban:

<https://sites.google.com/a/tnau.ac.in/nprc-vamban/varieties/redgram-varieties>

For more details on cultivation and varieties of red gram grown across India, please visit:

<https://agritech.tnau.ac.in/agriculture/CropProduction/Pulses/pulses redgram.html>.

Red gram and its products (HS Codes)

HS Code	Description
07136000	Pigeon peas (Cajanus cajan)
07139010	Other dried and shelled leguminous vegetables, split
07139090	Other of hdg. 071390
07089000	Other leguminous vegetable w/n shelled, fresh/chilled`

Other value added products of Pigeon Pea

As per United States Department of Agriculture, value added products are the products

- With different physical state (pigeon pea flour)
- That have different production techniques (organic)
- Preserved for longer shelf life (canned, dehydrated, frozen)

- **Pigeon Pea flour:** It is a change in the physical state of pigeon peas. The flour is economic and nutritional and can be used as a substitute to wheat. The product is in its nascent stage and there are studies that show that pigeon pea flour can produce protein-rich breads/ biscuits with improved essential amino acids. Pigeon pea flour can help the nations meet the second sustainability development goal of zero hunger. Researchers are studying the possibility of producing pasta with this flour as well.



- **Organic Pigeon Peas:** Pigeon peas produced as per the organic farming standards. Organic foods are generally seen by the customers as an unadulterated food that is grown in safer and cleaner environment. It is considered of high grade for consumption.

- **Canned Pigeon Peas:** Mature green pigeon peas are canned for export purpose. Canning improves the shelf life of the pigeon peas and preserve the taste of the product as well.





- **Dehydrates Pigeon Peas:** The shelf life of pigeon peas can be increased by dehydrating the product. Removing moisture from the food prevents it from microorganisms, and promotes consumption not just in the season, but throughout the year.

- **Frozen Pigeon Peas:** Freezing of the food preserves the greatest quantity of nutrients. By reducing the temperature of storage, we can prevent the growth of microorganisms leading to higher shelf life. Pigeon pea pods that are at the peak of their texture and flavour should be picked and frozen for consumption at a later stage. Delay in the freezing process can lead to reduced nutrient content.



Processed Products

Processed products are obtained due to conduction of series of mechanical/ chemical actions. The food has been altered by these actions. We could find three processed foods of pigeon pea, but the world trade data of these products is not available on the internet.

1. **Organic Tur Dal:** Pigeon peas are processed through various techniques to produce tur dal. It is also known as pigeon pea split dal with HSN code 07319090.
2. **Split Tur Dal:** Another processed product of pigeon peas with US - HSN code 07319010
3. **Probiotics:** With more people switching to vegan diets, there is an increased demand of non-dairy probiotic products in the market. The non-dairy probiotic products will also cater to lactose intolerant customer segment. Pigeon peas are being studied for production of pro-biotics and the initial studies find pigeon peas fit for commercial production of probiotics. This can be another processed food suitable for export.

Export Specifications of Tur Dal

Moisture content	13% Max
Foreign Matters	1% Max
Foreign Beans	0.5% Max
Weevilled Seeds	3% Max
Damaged Otherwise	3% Min

Packaging size

Packing	Available Size
PP Bag	15 & 25 Kg
For Human Consumption	1×20Ft Container

SPS measures for Red Gram

General requirements Red gram are:

- The dried mature grains of *Cajanus cajan* Linn only fit into the said category.
- They should be free from substances that render them unfit for human or animal consumption or processing into or the utilization thereof as food or feed.
- Should be free from abnormal flavors, sour, musty or other undesirable odour, discolouration, and obnoxious smell.
- Free from fungi, micro-organisms, substances originating from micro-organisms, or other poisonous substances in amounts that may constitute a hazard to human health. (Referenced from East African SPS standards, North American Standards)
- Should be free from certain pests, bugs based on origin country and these declarations are to be incorporated into SPS Certificate.

Contaminant

- Red gram should comply with pesticide residue limits established by the Codex Alimentarius Commission. Some countries have different limits, those can be found in respective countries trade sites.
- Other contaminants maximum limits established in CODEX STAN 193 need to be adhered to.
- In the case of red gram, Fumigation is needed by many countries, so we need to ensure that the cargo is treated properly by the licensed PCO. The export agricultural cargo/produce is fumigated by Plant Quarantine Institute or an authorized Private Pest Control Operators (PCOs).
- Once this is done Certificate will be issued.

Certain Special Conditions of Import imposed by some countries

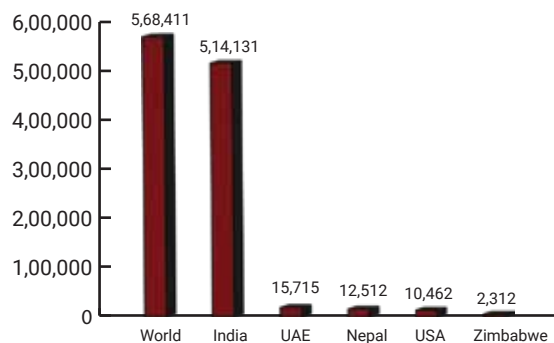
- Free from soil contamination,
- Fumigation by methyl bromide at 32 g per cubic meter for 24hrs at 21oC or another equivalent treatment approved by the government of importing countries has to be carried out and the treatment needs to be endorsed on the phytosanitary certificate issued at the country of origin or re-export.

World Exports and Imports

071360: Dried, shelled pigeon peas "Cajanus cajan", whether or not skinned or split.

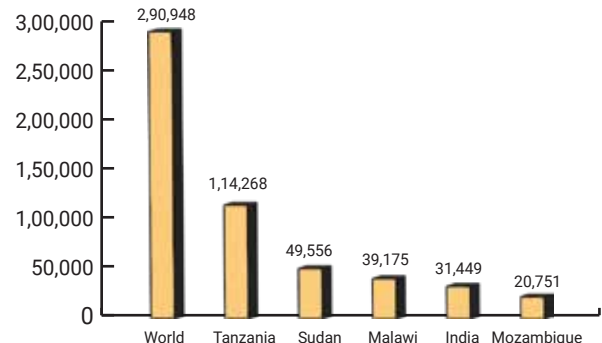
2021 Global Imports 568,411 USD.

No.	Country	Value USD
1	India	514,131
2	UAE	15,715
3	Nepal	12,512
4	USA	10,462
5	Zimbabwe	2,312



2021 Global Exports 290,948 USD.

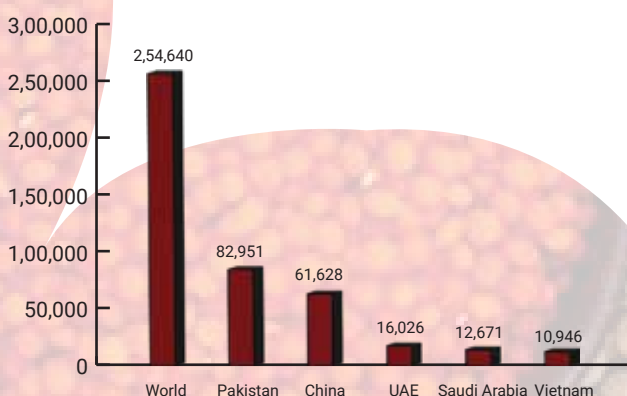
No.	Country	Value USD
1	Tanzania	114,268
2	Sudan	49,556
3	Malawi	39,175
4	India	31,449
5	Mozambique	20,751



071390: Dried, shelled leguminous vegetables, whether or not skinned or split (excluding peas, chickpeas, beans, lentils, broad beans, horse beans and pigeon peas).

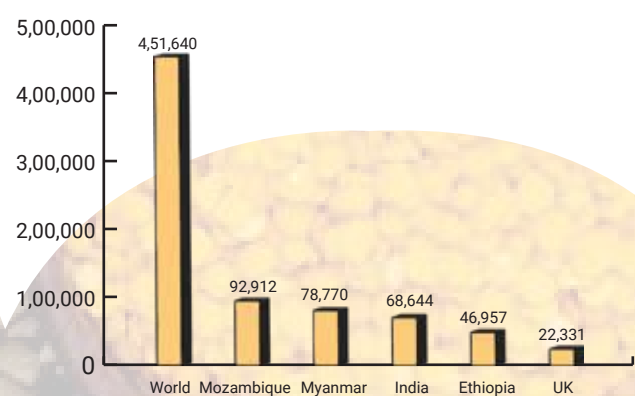
2021 Global Imports 254,640 USD.

No.	Country	Value USD
1	Pakistan	82,951
2	China	61,628
3	UAE	16,026
4	Saudi Arabia	12,671
5	Vietnam	10,946



2021 Global Exports 451,640 USD.

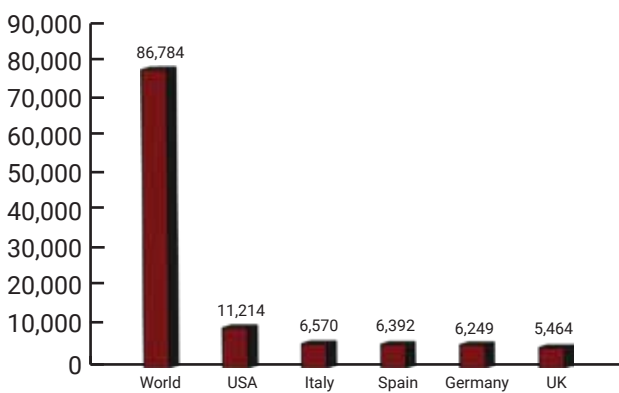
No.	Country	Value USD
1	Mozambique	92,912
2	Myanmar	78,770
3	India	68,644
4	Ethiopia	46,957
5	United Kingdom	22,331



070890: Fresh or chilled leguminous vegetables, shelled or unshelled (excluding peas "Pisum sativum" and beans "Vigna spp., Phaseolus spp").

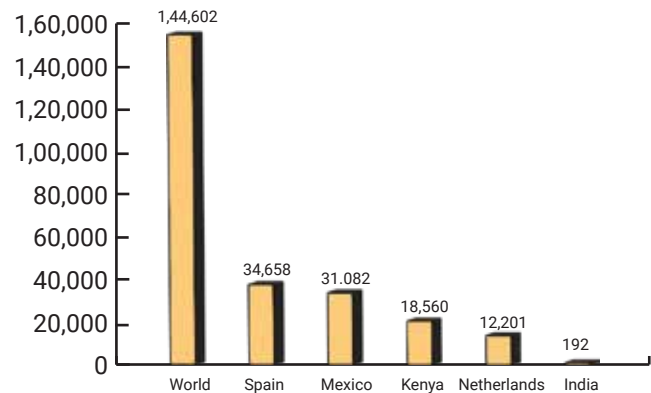
2021 Global Imports 86,784 USD.

No.	Country	Value USD
1	USA	11,214
2	Italy	6,570
3	Spain	6,392
4	Germany	6,249
5	UK	5,464



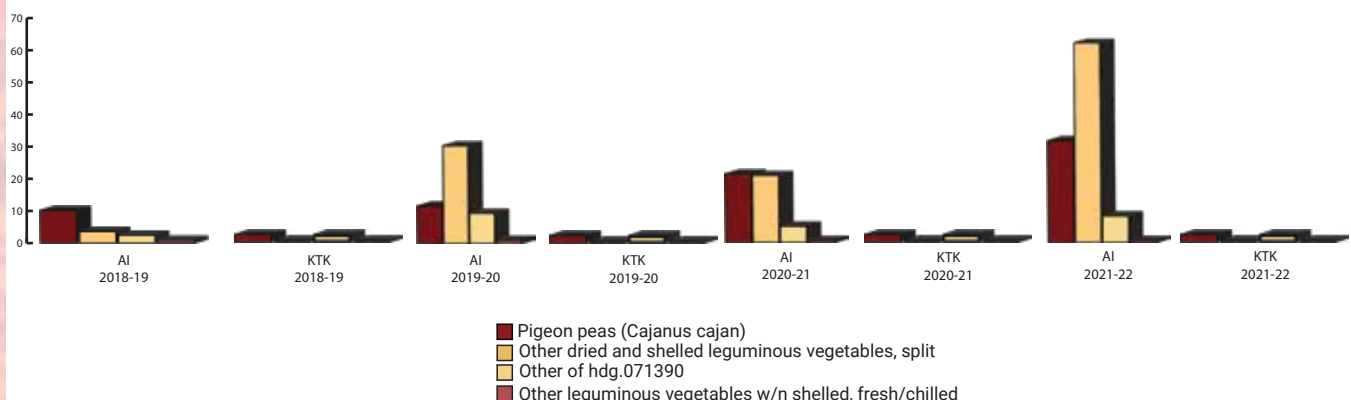
2021 Global Exports 144,602 USD.

No.	Country	Value USD
1	Spain	34,658
2	Mexico	31,082
3	Kenya	18,560
4	Netherlands	12,201
41	India	192



Pigeon peas Export Performance of last 4 years: Value in Mn USD

No.	HS Code	Description	2018-19		2019-20		2020-21		2021-22		Major Importing Countries
			AI	KTK	AI	KTK	AI	KTK	AI	KTK	
1	07136000	Pigeon peas (Cajanus cajan)	9.43	0.02	12.84	0.02	23.89	0.03	35.20	0.044	UAE, USA and Canada
2	07139010	Other dried and shelled leguminous vegetables, split	3.10	0.00	34.32	0.00	23.35	0.04	69.52	0.00	China, USA and Switzerland
3	07139090	Other of hdg. 071390	1.90	0.02	10.30	0.01	5.31	0.01	8.82	0.06	China, USA and Nepal
4	07089000	Other leguminous vegetable w/n shelled, fresh/chilled	0.43	0.00	0.59	0.01	0.24	0.00	0.17	0.00	Nepal, Canada and UK

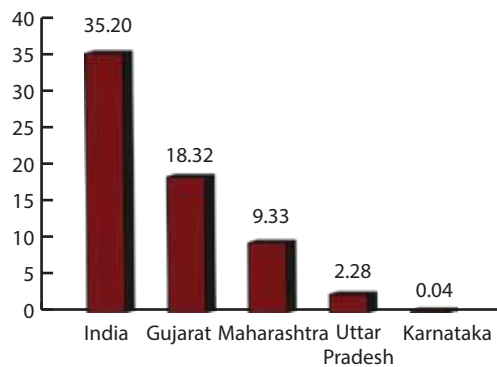


Export Analysis of Red gram and its products: 2021-22* (Karnataka)

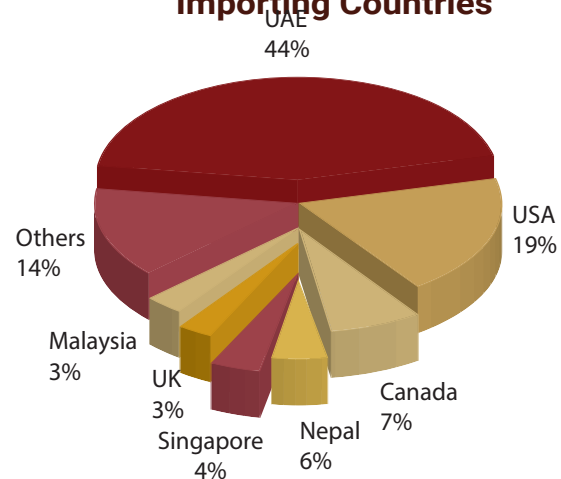
07136000: Pigeon peas "Cajanus cajan"

No.	State	Value USD Mn	Destination
	India	35.20	UAE, USA, Canada (+84)
1	Gujarat	18.32	UAE, USA, Canada (+55)
2	Maharashtra	9.33	UAE, Canada, UK (+46)
3	Uttar Pradesh	2.28	USA, Nepal, Singapore (+9)
12	Karnataka	0.04	UK, Singapore, Australia (+11)

Exporting States



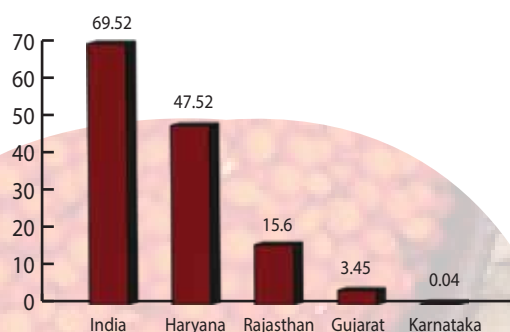
Importing Countries



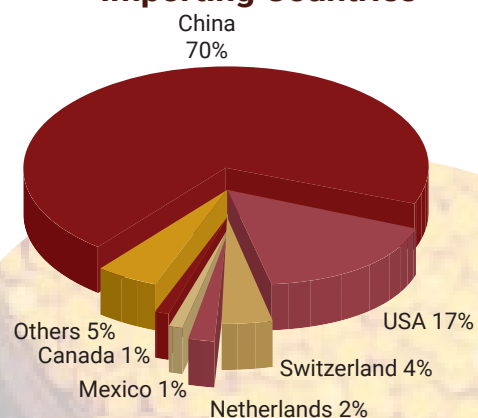
07139010: Other dried and shelled leguminous vegetables, split.

No.	State	Value USD Mn	Destination
	India	69.52	China, USA, Switzerland (+62)
1	Haryana	47.52	China, USA, Netherlands (+4)
2	Rajasthan	15.60	USA, China, Switzerland (+7)
3	Gujarat	3.45	China, USA, Mexico (+26)
0	Karnataka	0.04	Australia and Baharain

Exporting States



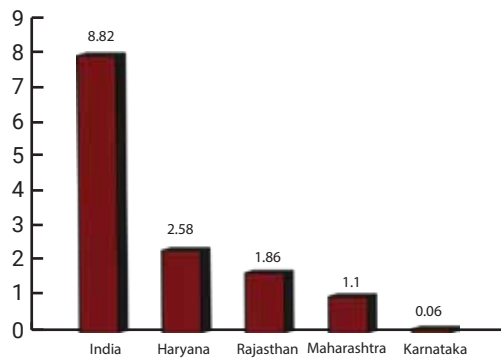
Importing Countries



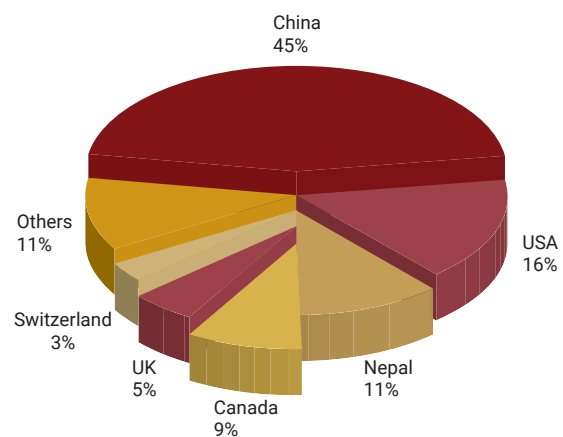
07139090: Other of hdg. 071390

No.	State	Value USD Mn	Destination
	India	8.82	China, USA, Nepal (+75)
1	Haryana	2.58	China, USA, Nepal (+4)
2	Rajasthan	1.86	China, USA, Switzerland (+12)
3	Maharashtra	1.10	Canada, USA, UK (+29)
11	Karnataka	0.06	UAE, Canada, Maldives (+2)

Exporting States



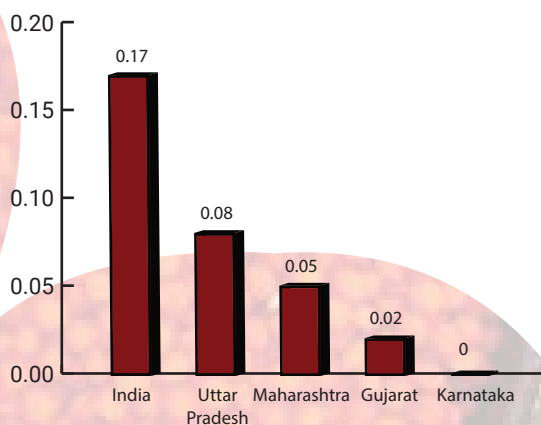
Importing Countries



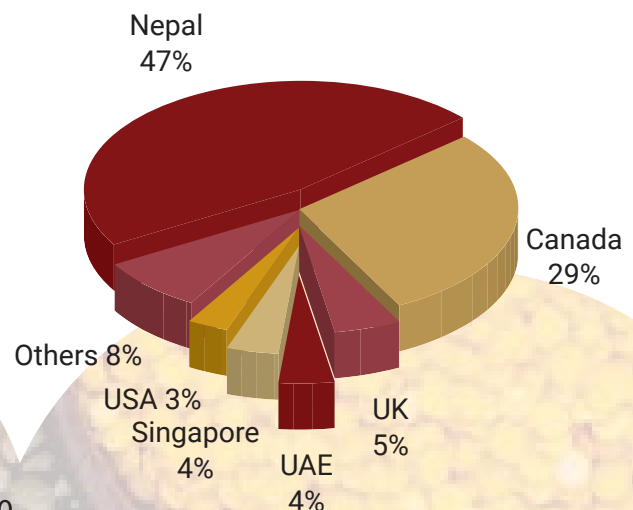
07089000: Other leguminous vegetable w/n shelled, fresh/chilled.

No.	State	Value USD Mn	Destination
	India	0.17	Nepal, Canada, UK (+15)
1	Uttar Pradesh	0.08	Nepal, UAE
2	Maharashtra	0.05	Canada, UAE, USA (+3)
3	Gujarat	0.02	Canada, South Africa, Seychelles (+5)
6	Karnataka	0.00	UAE

Exporting States



Importing Countries



Production Data

India is the largest producer (25% of global production), consumer (27% of world consumption) and importer (14%) of pulses in the world. The total pulse production of the country was 25.72 million tonnes in 2020-21 including Red gram (4.41 million tonnes). Source: FAO.

Production data state-wise

Sl. No.	State	2016-17	2017-18	2018-19	2019-20	2020-21*
1	Madhya Pradesh	6291.29	8111.58	6045.41	4108.41	4364.74
2	Rajasthan	3181.18	3405.37	3759.38	4497.13	4821.84
3	Maharashtra	3768.06	3347.76	2682.54	3736.00	4224.87
4	Uttar Pradesh	2184.39	2199.99	2408.01	2447.32	2621.15
5	Karnataka	1737.90	1951.21	1773.86	2155.89	2170.89

*As per 3rd Advance Estimates.

(Source: <https://pib.gov.in/PressReleaselframePage.aspx?PRID=1741992>)

GI tagged Tur Dal

Kalaburagi, a major producer of tur dal (red gram) and is known as the 'tur dal bowl' of Karnataka. Tur dal is cultivated in around 70% of the agricultural land in the district and the red gram grown in Kalaburagi is commonly known as the 'Gulbarga tur dal'. It has high calcium and potassium content – mainly attributable to the mineral-rich black soil of the district. Owing to its superior taste and aroma, the Gulbarga tur dal has a GI tag. This variety also takes less time to cook compared to the red gram grown in other regions.

Milling technologies available

1. Small scale mill:

<https://cftri.res.in/kweb/technologies/FM/vdm.pdf>

2. Mini mill:

<https://cftri.res.in/kweb/technologies/FM/mvd.pdf>

3. Analytical report on CFTRI Mill:

http://researchjournal.co.in/upload/assignments/10_239-245.pdf

Action plan

1. Focus on quality development and yield improvement in existing tur dal varieties of India.
2. Import best agricultural practices and bring in global varieties germ plasm to improvise local varieties through agricultural and horticultural universities.
3. Orientation and Facilitation must be prioritized to bridge the gap between research institutes, academia, industries and farmers to facilitate exports.
4. A local APEDA testing lab and pack house in red gram growing areas, helps growers to understand quality standards and thereby tap international markets.
5. Create awareness about the new products which have international market and transfer knowledge about required technology through CFTRI and IIFPT, so as to produce value added products.
6. Though India is the largest exporter of Red gram from Asian continent, Karnataka is not among the top ten exporting states and holds no position in processed products like canned, dehydrated, frozen fresh red gram and tur dal flour. Hence there is a need to encourage first generation entrepreneurs and start-ups to establish themselves in processing sector to produce value added products and also to divert the surplus produce in turn avoiding the financial losses.
7. State and Central governments must sponsor financial schemes/ subsidies to aid capacity building in setting up of industries and encourage skill development programmes for the youth.



Opportunities in Red gram and related products

1. Though Red gram has a huge international market, Karnataka is not contributing much to this sector. Apart from dry seed milling, not much has been done in processed products of dal and there is a lot of demand in these sectors yet unexplored by Indian growers.
2. The value-added products not only provide employment to a larger group locally but also help the industries to establish themselves in exports.
3. Setting up of a dedicated testing lab, either stationary or mobile helps local growers.
4. Dedicated pack house in North Karnataka is the need of the hour.
5. Processing industries for canned, frozen fresh red gram and dal flour will easily fetch revenues from local as well as international market.

Responsibility Matrix

No.	Actions	Responsibility
1	Research on best global varieties.	Agricultural and Horticultural Universities, Karnataka.
2	Delegation of progressive farmers to Mexico, Spain and Netherlands.	Agriculture department, Karnataka.
3	Farmer Sensitization programs on available opportunities.	Agriculture department, Karnataka.
4	Inviting Exporters/entrepreneurs to Investors meet for contract farming and processing industries	Agriculture Department and DIC Districts.
5	Farmer database and aggregation of the produce.	Agriculture department and related FPOs
6	Processing technology and Value Addition Handholding	Agriculture department through CFTRI, Mysore and IIFPT, Thanjavur.
7	Export training, orientation and Market Intelligence	VTPC Karnataka

Regulators and Service providers

No.	Organization	Service	Contact
1	APEDA	RCMC, Market survey and assistance.	1st Floor, Beeja Bhavan, Bellary Rd, Hebbal, Bengaluru - 560024. E: apedabl@apeda.gov.in
2	KAPPEC	PMFME Scheme	17, Richmond Rd, Shanthala Nagar, Richmond Town, Bengaluru - 560025. E: kappec1996@gmail.com
3	DGFT	IEC, Customs and ICEGATE	6th floor, Kendriya Sadan, C & E Wing, 17th main, Koramangala 2nd Block, Koramangala, Bengaluru - 560034 E: bangalore-dgft@nic.in
4	Plant Quarantine	Phyto Sanitary certification (SPS)	Hebbal-Boopasandra Road HA Farm Post, Bengaluru - 560024. E: dd-pqfsb-ka@nic.in
5	BIAL Cool Port	Freight and Flight	KIAL Road, Devanahalli, Bengaluru -560300 W: http://www.aisats.in

Tur dal processing technologies available

Pigeon pea milling basically includes following steps – cleaning and grading of grains, pitting, pre-treatment, tempering, drying, de-husking and splitting and polishing.

Pigeon pea are the most difficult to mill and direct milling of pigeon peas without the pre- treatment results in lower dal recovery. Following are the various pre-treatment techniques available:

- **Water-spray treatment:** Pigeon peas are soaked in the water to loosen the binding action between seed and cotyledons. Increased moisture levels results in softer seed coat. In large mills, the water is sprinkled or sprayed on pigeon peas, hence the name of the treatment. Alternate wetting and drying technique is also used in place of water-spray treatment.
- **Oil Treatment:** This treatment can take several days. The oil is allowed to penetrate between seed and cotyledons. The mild heat of the sun results in loose bond between the layers.

- **Chemical treatment:** Sodium bicarbonate, sodium carbonate, calcium hydroxide, and sodium chloride are widely used chemicals for pre-treatment. These chemicals are the most effective, resulting in highest dal recovery. But the nutrient content gets reduced because of this pre-treatment.
- **Heating treatment:** Pigeon peas can be heated by roasting on sand or in a hot chamber at temperatures between 100 degree Celsius and 140 degree Celsius. This is a quick and effective way of loosening the bond between layers. The treatment can take as low as 10 mins.

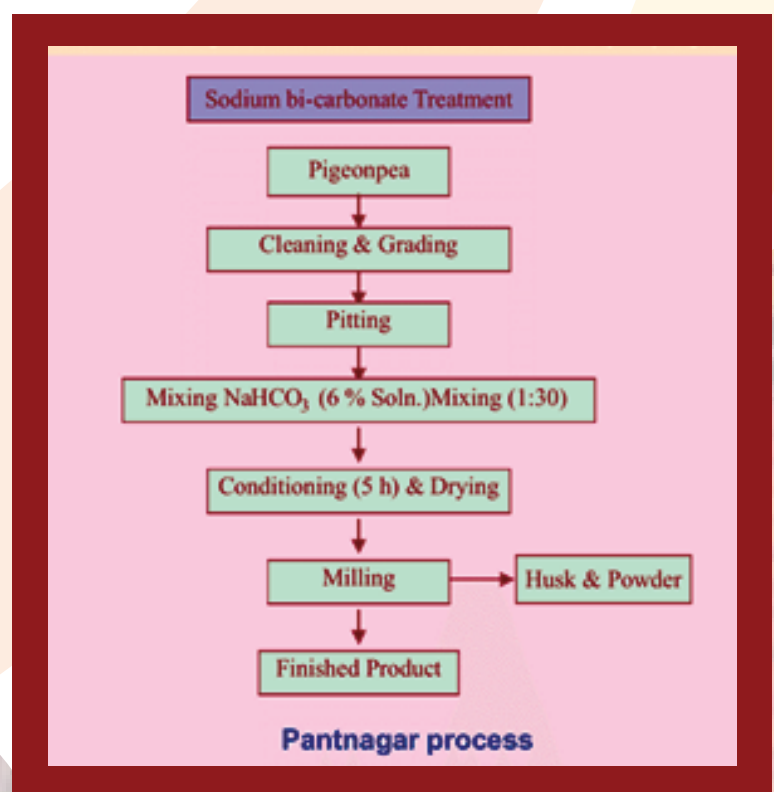
Traditionally, wet and dry treatment techniques are used as pre-treatment. Several new techniques have been developed that use the combination of above mentioned pre-treatment techniques to get the highest dal recovery. It is a proud fact that all these have been developed in India.

Pantnagar process using Enzymatic treatment

In this process, the pigeon pea grains are first treated with enzyme and then pre-treated with different combinations of water-spray treatment and heating treatment. The grains treated with enzymes and water had dal recovery close to 90%. Enzyme treatment is said to improve the digestibility of protein in dal and significantly reduces cooking time.

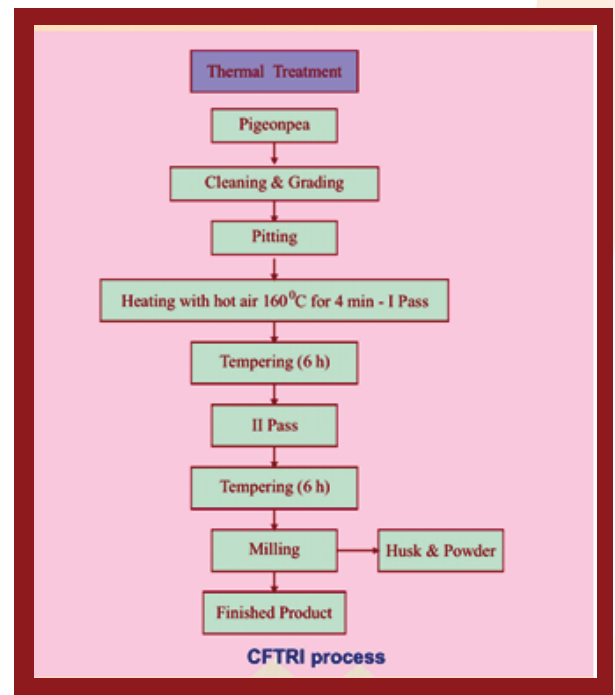
Pantnagar Process using Chemical Treatment

Pigeon pea is pre-treated with 10% sodium bicarbonate solution. These grains are then heated at 30 °C for 5 hours, followed by drying directly under the sun. The process uses a combination of chemical and heating treatment. This process can get dal recovery of close to 80%. The dal recovery is quite high, but the chemical treatment might make the husk unfit for cattle feeding.



CFTRI process

The process is developed by Central Food Technological Research Institute based in Mumbai. This process overcomes the limitation of the previous process by making the milling weather independent. Heating and drying process is carried out indoors and replaces the dependency on sun, making it weather independent. This process can give as much as 80% dal recovery. The barriers in adoption of this technique by mills are the requirements of continuous supply of electricity and high electricity consumption. The overall cost of machinery is quite high, making it less preferable. This technique cannot be integrated with existing traditional machinery.



CIAE Process

This process was developed by Central Institute of Agricultural Engineering situated in Bhopal.

Grains are soaked in water for 25-30 minutes at room temperature to produce 35% moisture content and then dried to bring the moisture content down to 10%. The process uses a combination of water-spray and heat treatment and these dried grains are milled to produce split dal. With this method, the use of edible oil is eliminated from the milling process and dal recovery was close to 75%. Weather conditions play a role in this process.

